Junseo Lee

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Harvard Quantum Initiative, Goel Quantum Science and Engineering Building,
60 Oxford St, Cambridge, MA 02138, United States (Office: TBD)

Education

Harvard University — Cambridge, MA
Doctor of Philosophy (Ph.D.), Quantum Science and Engineering — (Starting in Fall 2026)

- Affiliated with the Harvard Quantum Initiative and the Theory of Computation Group

Yonsei University — Seoul, Korea
Bachelor of Science (B.S.), Electrical and Electronic Engineering — Mar. 2019 – Feb. 2023

- Fully funded by the Hyundai Motor Chung Mong-Koo (CMK) Scholarship in Intelligent Information Technology
- Thesis: *Combinatorial Designs in Information Theory*

Chungnam Science High School — Gongju, Korea
Concentration in Mathematics, Early Honors Graduate — Mar. 2017 – Dec. 2018

Research Interests

Quantum Learning Theory, Quantum Complexity Theory, Quantum Topology, Theoretical Computer Science

Research Experience

Seoul National University — Seoul, Korea
Research Fellow, Institute of Computer Technology (Host: Dr. Kabgyun Jeong) — Apr. 2026 – Present

- Funded by the National Research Foundation of Korea (NRF), the Institute for Information and Communications Technology Planning and Evaluation (IITP), and the Electronics and Telecommunications Research Institute (ETRI)

Research Affiliate, Research Institute of Mathematics — Jan. 2023 – Mar. 2026
Research Assistant, Department of Mathematical Sciences — Mar. 2020 – Dec. 2022

Republic of Korea Army — Seoul, Korea
Research Scientist (Technical Research Personnel, Mandatory National Military Service) — Mar. 2023 – Mar. 2026

Publications (Google Scholar Profile)

*Equal contribution. †Authors listed alphabetically.

Preprints

(17) Kabgyun Jeong, Donghwa Ji, **Junseo Lee**†, Adam Sawicki, and Oskar Słowik, "Explicit bounds on polylogarithmic spectral gap decay in unitary channels," (2026). In preparation.

(16) **Junseo Lee**† and Myeongjin Shin, "Optimal certification of constant-local Hamiltonians," (2025). arXiv:2512.09778 [quant-ph].
► Featured in the Monthly News of Property Testing Review (December 2025).

(15) Marco Fanizza, Vishnu Iyer, **Junseo Lee**†, Antonio A. Mele, and Francesco A. Mele, "Efficient learning of bosonic Gaussian unitaries," (2025). arXiv:2510.05531 [quant-ph].
► Contributed talk, **QIP 2026**.

(14) Nhat A. Nghiem, **Junseo Lee**, and Tzu-Chieh Wei, "Hybrid quantum-classical framework for Betti number estimation with applications to topological data analysis," (2025). arXiv:2508.01516 [quant-ph].

(13) Kartik Anand, Kabgyun Jeong, and **Junseo Lee**†, "Collapses in quantum-classical probabilistically checkable proofs and the quantum polynomial hierarchy," (2025). arXiv:2506.19792 [quant-ph].

(12) **Junseo Lee**† and Nhat A. Nghiem, "New aspects of quantum topological data analysis: Betti number estimation, and testing and tracking of homology and cohomology classes," (2025). arXiv:2506.01432 [quant-ph].

Journal Articles

(11) Donghwa Ji, **Junseo Lee**, Myeongjin Shin, IlKwon Sohn, and Kabgyun Jeong, "Bounding quantum uncommon information with quantum neural estimators," *Quantum Science and Technology* **11**, 015001 (2026). DOI: 10.1088/2058-9565/ae18f4.

(10) Myeongjin Shin*, **Junseo Lee***, Seungwoo Lee, and Kabgyun Jeong, "Resource-efficient algorithm for estimating the trace of quantum state powers," *Quantum* **9**, 1832 (2025). DOI: 10.22331/q-2025-08-27-1832.
► Featured in the Monthly News of Property Testing Review (August 2025).

(9) Mingyu Lee, Myeongjin Shin, **Junseo Lee**, and Kabgyun Jeong, "Mutual information maximizing quantum generative adversarial networks," *Scientific Reports* **15**, 32835 (2025). DOI: 10.1038/s41598-025-18476-y.

(8) Myeongjin Shin*, Seungwoo Lee*, **Junseo Lee***, Donghwa Ji, Hyeonjun Yeo, and Kabgyun Jeong, "Disentanglement provides a unified estimation for quantum entropies and distance measures," *Physical Review A* **110**, 062418 (2024). DOI: 10.1103/PhysRevA.110.062418.

(7) Myeongjin Shin, **Junseo Lee**, and Kabgyun Jeong, "Estimating quantum mutual information through a quantum neural network," *Quantum Information Processing* **23**, 57 (2024). DOI: 10.1007/s11128-023-04253-1.

(6) **Junseo Lee** and Kabgyun Jeong, "Quantum Rényi entropy functionals for bosonic gaussian systems," *Physics Letters A* **490**, 129183 (2023). DOI: 10.1016/j.physleta.2023.129183
► PLA Special Issue, Foundations and applications of Quantum Optics (2024).

(5) **Junseo Lee**, Hyeonjun Yeo, and Kabgyun Jeong, "Weighted p-Rényi entropy power inequality: Information theory to quantum Shannon theory," *International Journal of Theoretical Physics* **62**, 253 (2023). DOI: 10.1007/s10773-023-05512-8

(4) **Junseo Lee** and Kabgyun Jeong, "High-dimensional private quantum channels and regular polytopes," *Communications in Physics* **31**, 189 (2021). DOI: 10.15625/0868-3166/15762
► Third Prize, Undergraduate Research Exhibition, Korean Physical Society (2021).

(3) Kabgyun Jeong, **Junseo Lee**, *et al.*, "Single qubit private quantum channels and 3-dimensional regular polyhedra," *New Physics: Sae Mulli* **68**, 232 (2018). DOI: 10.3938/NPSM.68.232
► Bronze Award, The Humantech Paper Award, Samsung Electronics (2018).

Book Chapters

(2) **Junseo Lee**, "Assessing Quantum Integer Factorization Performance with Shor's Algorithm," In: *Quantum Computing: A Journey into the Next Frontier of Information and Communication Security*, Edited by Mohammad Hammoudeh, Abdullah T. Alessa, Amro M. Sherbeeni, Clinton M. Firth, Abdullah S. Alessa, *CRC Press* (2024). DOI: 10.1201/9781003475286

Patents

(1) Kabgyun Jeong, Myeongjin Shin, and **Junseo Lee**, "Method for estimating quantum mutual information through a quantum neural network," *Korea Patent* Open No. 10-2026-0009068 (2024). DOI: 10.8080/1020240091151

Professional Activities

Journal Reviewer

- **American Physical Society:** Physical Review Letters, PRX Quantum, Physical Review Research, Physical Review Applied, Physical Review A
- **Nature Research:** npj Quantum Information
- **Institute of Electrical and Electronics Engineers:** IEEE Transactions on Information Theory
- **Others:** Quantum, Physics Letters A, Annalen der Physik

Conference Reviewer

- Theory of Quantum Computation, Communication and Cryptography (TQC), 2026
- Quantum Computing Theory in Practice (QCTiP), 2026
- Quantum Techniques in Machine Learning (QTML), 2025

Community Service

- *Poster Session Judge*, National Undergraduate Quantum Conference, Seoul National University, 2026
- *Creator and Maintainer*, Quantum Learning Theory Zoo, curated repository of quantum learning theory papers, 2025 – Present
- *Selection Committee*, Quantum Internship Program, National Information Society Agency and Korea Quantum Industry Center, 2024 – 2025
- *Co-organizer*, SNU Quantum Information Theory Seminar, 2024 – 2025
- *Co-organizer*, Quantum AI Hackathon, jointly organized by Kakao Enterprise Corp. and Jeonju University, 2025
- *Facilitator (Mentor)*, Mathematics Section, Korea Scholar's Conference for Youth (KSCY), Yonsei University, 2019

Selected Honors and Awards

Funding and Fellowships

- *Academic Travel Grant*, Hyundai Motor CMK Foundation, 2022
- *Hyundai Motor CMK Scholarship in Intelligent Information Technology*, full tuition and living stipend, 2021–2022
- *Teaching Fellowship for Software Courses*, Yonsei University, 2021–2022
- *JH Science Scholarship*, 2017

Additional Honors and Awards

- *High Honor Student*, Yonsei University, 2022
- *Selected Paper Award*, Finance and Economics Contest, DB Group, 2022
- *Best Tutor Award*, Yonsei University, 2021–2022
- *Third Prize*, Undergraduate Research Exhibition, Korean Physical Society, 2021
- *Honor Student*, Yonsei University, 2020–2021
- *Bronze Award*, The Humantech Paper Award, Samsung Electronics, 2018
- *Best Translator Award*, NAVER Connect Foundation and Khan Academy, 2018
- *National Honorable Mention and Regional Gold Award*, Korean Olympiad in Informatics, 2016

Teaching Experience

Quantum Information Science Club Association (2025–2026)
(Teaching materials are available at: harris-junseo-lee.github.io/teaching/)

- *Invited Lecturer*, Quantum Learning Theory for Bosonic and Fermionic Systems, Winter 2026
- *Invited Lecturer*, Quantum Complexity Reading Group, Fall 2025
- *Invited Lecturer*, Quantum Learning and Complexity Theory, Summer 2025

University–Industry Research Internship (2024–2025)

- *Instructor*, AAA558/AAA559: College of Informatics Internship, Korea University (Graduate Course), Fall 2025
- *Instructor*, SW4343: Software Field Placement 1, Korea Aerospace University, Fall 2024

Yonsei University (2021–2022)

- *Teaching Assistant*, YCS1009: Change the World through Programming, Fall 2022
- *Teaching Assistant*, YCS1002: Software Programming, Fall 2022
- *Teaching Assistant*, EEE1108: Engineering Information Processing, Fall 2021
- *Course Tutor*, MAT2016: Engineering Mathematics 3, Spring 2022 [Best Tutor Award]

- *Course Tutor*, MAT1012: Engineering Mathematics 2, Fall 2021 [Best Tutor Award]

Selected Talks

*Online talk.

Research Talks

"Optimal certification of constant-local Hamiltonians"

- *Invited talk*, Quantum Software Lab Seminar, University of Edinburgh, Mar. 2026*

"Efficient learning of bosonic Gaussian unitaries"

- *Invited talk*, Annual Meeting of the Quantum Information Society of Korea, Feb. 2026
- *Invited talk*, N^3etFraST Workshop, Nov. 2025
- *Invited talk*, Yonsei Quantum Data Science & AI Lab Seminar, Nov. 2025
- *Contributed talk*, **QIP 2026**, Jan. 2026 (Presented under the title "Efficient Learning Algorithms for Structured Bosonic and Fermionic Unitary Operators", as a merged submission with `arXiv:2504.11318`.)

"New aspects of quantum topological data analysis"

- *Invited talk*, KISTI-SNU Joint Workshop, Jun. 2025

"Resource-efficient algorithm for estimating the trace of quantum state powers"

- *Invited talk*, Electronics & Telecommunications Research Institute, Dec. 2024
- *Invited talk*, SNU Quantum Information Theory Seminar, Dec. 2024*
- *Invited talk*, IBM-Yonsei Qiskit Fall Fest, Nov. 2024*
- *Contributed talk*, Annual Meeting of Korean Mathematical Society, Oct. 2024
- *Poster*, QIP 2025, Feb. 2025

"Mutual information maximizing quantum generative adversarial network"

- *Invited talk*, Triangle Quantum Computing Seminar, NC State University Quantum Initiative, Nov. 2023*

"Estimating quantum mutual information through a quantum neural network"

- *Invited talk*, CS Katha Barta, National Institute of Science Education and Research Bhubaneswar, Aug. 2023*

"Quantum Rényi entropy functionals for bosonic Gaussian systems"

- *Poster*, QIP 2022, Mar. 2022

"High-dimensional private quantum channels and regular polytopes"

- *Invited talk*, KISTI-KU-SNU Joint Workshop, Sep. 2023*
- *Invited talk*, SNU Quantum Information Theory Seminar, Aug. 2021*
- *Contributed talk*, Winter Meeting of the Optical Society of Korea, Feb. 2022
- *Contributed talk*, Fall Meeting of the Korean Physical Society, Oct. 2021*
- *Poster*, QIP 2022, Mar. 2022

Tutorials and Public Lectures

"Learning theory in ∞-dimensional quantum systems"

- *Invited talk*, Team QST Summer Workshop, Seoul National University, Aug. 2025

"Introduction to quantum machine learning"

- *Invited talk*, AWS Korea Healthcare & Research Team Seminar, Mar. 2025

"Topics in theoretical quantum computer science"

- *Invited talk*, Shinil High School, Aug. 2024

"Quantum machine learning models for drug library generation"

- *Invited talk*, Yonsei Quantum Computing and Monte Carlo Workshop, Aug. 2024

"$\mathsf{QMA} \stackrel{?}{=} \mathsf{NP}$: The NLTS theorem and the quantum PCP conjecture"

- *Invited talk*, SNU Center for Quantum Network's Channel Capacity Summer Workshop, Jul. 2024

"Minimal data may be sufficient for quantum artificial intelligence"

- *Invited talk*, SNU Department of Mathematical Sciences Seminar, Jun. 2023*

Skills and Technical Experience

Programming Languages: Proficient in C, C++ (Informatics Olympiad), and Python; experienced with Java.

Quantum Software: Proficient in PennyLane and IBM Qiskit (certified); experienced with $Q^{\#}$ and PyZX.

- *IBM Certified Associate Developer*, Quantum Computation using Qiskit, 2023
- *Advanced Achievement*, IBM Quantum Spring Challenge, 2023
- *Advanced Achievement*, Xanadu QHack Coding Challenges, 2023